

AI & Chatbot Development Roadmap (For Beginners)

PHASE 0: Prerequisites Check (1 week)

- Python Basics (variables, loops, functions, lists, dictionaries, file handling, pip).

PHASE 1: Math for AI (2–3 weeks)

- Linear Algebra: Vectors, Matrices, Dot Product, Matrix Multiplication.
- Calculus: Derivatives, Gradients, Chain Rule.
- Probability & Statistics: Mean, Variance, Normal Distribution, Bayes' Theorem.
- Optimization: Loss functions, Gradient Descent.

PHASE 2: Python for AI (2–3 weeks)

- NumPy – Array operations.
- Pandas – Data manipulation.
- Matplotlib / Seaborn – Visualization.
- Jupyter Notebooks – Interactive coding.

PHASE 3: Machine Learning Fundamentals (4–6 weeks)

- Supervised vs Unsupervised Learning.
- Model training, overfitting, evaluation metrics.
- Key Algorithms: Linear/Logistic Regression, Decision Trees, KNN, K-Means.
- Scikit-Learn usage.

PHASE 4: Deep Learning & Neural Networks (4–6 weeks)

- Neural Networks, Layers, Activation Functions.
- Backpropagation, CNNs, RNNs, LSTMs.
- Use PyTorch or TensorFlow to build models.

PHASE 5: Natural Language Processing (4 weeks)

- Text Preprocessing: Tokenization, Lemmatization.
- Embeddings: Word2Vec, GloVe.
- Sequence Models: RNN, LSTM, Transformer.
- Pretrained Models: BERT, GPT.

PHASE 6: Build Your Own Chatbot (4–6 weeks)

- Traditional Chatbots: Intent Recognition, Rasa/Dialogflow.
- LLM-Based Chatbots: OpenAI API, GPT.
- Vector Databases & RAG: FAISS, Pinecone.
- LangChain, Streamlit, Gradio for interface.

PHASE 7: Deployment (2–3 weeks)

- Flask/FastAPI for API.
- Deploy with Vercel, Hugging Face, or Docker.

PHASE 8: Portfolio & Practice (Ongoing)

- Build projects, publish on GitHub.
- Contribute to open-source, join AI communities.

Estimated Timeline

- Phase 0: 1 week
- Phase 1: 2–3 weeks
- Phase 2: 2–3 weeks
- Phase 3: 4–6 weeks
- Phase 4: 4–6 weeks
- Phase 5: 4 weeks
- Phase 6: 4–6 weeks
- Phase 7: 2–3 weeks
- Phase 8: Ongoing